## * Non-linear Differential Equations Reducible to linear form :-

### Type - I :- Bernoulli's Equation:-

- The equation of the form $\frac{dy}{dx} + Py = Qy^n$ — (1)

where P & Q are functions of $x$ or constants, is a non-linear eq$^n$ known as Bernoulli's equation.

- This eq$^n$ can be made linear using the following method.
- Dividing eq$^n$ (1) by $y^n$,

$$\frac{1}{y^n}\frac{dy}{dx} + \frac{P}{y^{n-1}} = Q. \quad — (2)$$

Let, $\frac{1}{y^{n-1}} = v.$

Diff. w.r.t to $x$, we get

$$-(n-1)\, y^{-(n-1)-1} \frac{dy}{dx} = \frac{dv}{dx}$$

$$\frac{(1-n)}{y^n}\frac{dy}{dx} = \frac{dv}{dx}.$$

$$\therefore \frac{1}{y^n}\frac{dy}{dx} = \frac{1}{1-n}\frac{dv}{dx}.$$

Substituting this in eq$^n$ (2),

$$\frac{1}{1-n}\frac{dv}{dx} + Pv = Q.$$

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$\therefore \frac{dv}{dx} + (1-n) Pv = Q.$

- This eq$^{n}$ is linear in $v$ and can be solved using the method of linear differential eq$^{ns}$.

- Finally, substituting $v = \frac{1}{y^{n-1}}$, we get the sol$^{n}$ of eq$^{n}$ (1).

* Examples:-

(1) Solve $xy' = y^2 + y$.

⇒

The given D.E. can be written as

$x \cdot \frac{dy}{dx} - y = y^2$

- To make the co-efficient of $\frac{dy}{dx}$ unity, devide by $x$,

$\therefore \frac{dy}{dx} - \frac{1}{x} y = \frac{y^2}{x}$.

Comparing it to Bernoulli's eq$^{n}$, $\frac{dy}{dx} + Py = Qy^n$,

$P = -\frac{1}{x}$, Dividing both the sides by $y^2$,

$\frac{1}{y^2} \frac{dy}{dx} - \frac{y}{xy^2} = \frac{1}{x}$

$\therefore y^{-2} \frac{dy}{dx} - \frac{1}{x} y^{-1} = \frac{1}{x}$ ——— (1)

Let, $y^{-1} = v$

Diff. w.r.to $x$,

$-y^{-2} \frac{dy}{dx} = \frac{dv}{dx}$

From (1),

$-\frac{dv}{dx} - \frac{1}{x} v = \frac{1}{x}$

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$$\therefore \frac{dv}{dx} + \frac{1}{x} v = -\frac{1}{x} \quad \text{---} (2)$$

Which is linear eq$^n$ in $v$.

$$\text{I.F.} = e^{\int P\,dx} = e^{\int \frac{1}{x} dx} = e^{\log x} = x.$$

- The G.S. is

$$v(\text{I.F.}) = \int Q(\text{I.F.})\,dx + c$$

$$\therefore vx = \int \left(-\frac{1}{x}\right) x\,dx + c$$

$$\therefore vx = -x + c$$

$$\therefore \frac{1}{y} x = -x + c, \text{ where } c \text{ is an arbitrary constant.}$$

(2) Solve the Bernoulli's eq$^u$ $y' + y \sin x = e^{\cos x}$.

→

The given D.E. is of the form $\frac{dy}{dx} + Py = Q$.

Here, $P = \sin x$, $Q = e^{\cos x}$.

$$\text{I.F.} = e^{\int P\,dx} = e^{\int \sin x\,dx} = e^{-\cos x}.$$

∴ The G.S. is

$$y \cdot e^{-\cos x} = \int e^{\cos x} \cdot e^{-\cos x}\,dx + c$$

$$\therefore y e^{-\cos x} = x + c.$$

(3) $y^4 dx = \left(x^{-3/4} - y^3 x\right) dy$.

→

Rewritting the eq$^u$ as

$$\frac{dx}{dy} = \frac{x^{-3/4} - y^3 x}{y^4} = \frac{x^{-3/4}}{y^4} - \frac{y^3 x}{y^4}$$

$$\therefore \frac{dx}{dy} = \frac{1}{y^4} x^{-3/4} - \frac{1}{y} x$$

$$\therefore \frac{dx}{dy} + \frac{1}{y} x = \frac{1}{y^4} x^{-3/4}$$

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- The eqⁿ is in Bernoulli's form, where x is dep variable.

- Dividing the eqⁿ by $x^{-3/4}$,

$$\therefore x^{3/4}\frac{dx}{dy} + x^{7/4}\left(\frac{1}{y}\right) = \frac{1}{y^4} \quad \text{—— (1)}$$

Let, $x^{7/4} = v$

$$\therefore \frac{7}{4}x^{3/4}\frac{dx}{dy} = \frac{dv}{dy} \Rightarrow x^{3/4}\frac{dx}{dy} = \frac{4}{7}\frac{dv}{dy}$$

$\therefore$ From (1),

$$\frac{4}{7}\frac{dv}{dy} + \left(\frac{1}{y}\right)v = \frac{1}{y^4}$$

$$\therefore \frac{dv}{dy} + \left(\frac{7}{4y}\right)v = \frac{7}{4y^4} \quad \text{—— (2)}$$

- The eqⁿ is linear in v.

$$\therefore P = \frac{7}{4y}, \quad Q = \frac{7}{4y^4}$$

$$I.F. = e^{\int P\,dy} = e^{\int \frac{7}{4y}dy} = e^{\frac{7}{4}\log y} = e^{\log y^{7/4}} = y^{7/4}.$$

∴ G.S. of eqⁿ (2) is

$$v \cdot y^{7/4} = \int \frac{7}{4y^4}\cdot y^{7/4}\,dy + c$$

$$= \frac{7}{4}\int y^{-9/4}\,dy + c$$

$$= \frac{7}{4}\left[-\frac{4}{5}y^{-5/4}\right] + c$$

$$y^{7/4}\cdot v = -\frac{1}{5}y^{-5/4} + c$$

$$\therefore y^3 \cdot x^{7/4} = -\frac{7}{5} + cy^{5/4}.$$

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(4) Solve $\frac{dr}{d\theta} = r\tan\theta - \frac{r^2}{\cos\theta}$

→

Rewritting the eq$^n$, $\frac{dr}{d\theta} - r\tan\theta = -\frac{1}{\cos\theta} r^2$

- The eq$^n$ is in Bernoulli's form, where $r$ is dependent variable.

- Dividing by $r^2$,

$\frac{1}{r^2}\frac{dr}{d\theta} - \frac{1}{r}\tan\theta = -\frac{1}{\cos\theta}$ —— (1)

Let, $-\frac{1}{r} = v$

$\therefore \frac{1}{r^2}\frac{dr}{d\theta} = \frac{dv}{d\theta}$

From (1),

$\frac{dv}{d\theta} + v\tan\theta = -\frac{1}{\cos\theta}$ —— (2)

Which is linear in $v$.

$P = \tan\theta, \quad Q = -\frac{1}{\cos\theta}$.

$I.F. = e^{\int P d\theta} = e^{\int \tan\theta\, d\theta} = e^{\log\sec\theta} = \sec\theta$.

- The G.S. of eq$^n$ (2) is

$v \cdot \sec\theta = \int \sec\theta \cdot \left(\frac{-1}{\cos\theta}\right) d\theta + c$

$= \int -\sec^2\theta\, d\theta + c$

$= -\tan\theta + c$.

$\therefore \sec\theta\left(-\frac{1}{r}\right) = -\tan\theta + c$

$\therefore \frac{\sec\theta}{r} = \tan\theta - c$

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## * Type 2:- Riccati's eqⁿ:-

- The eqⁿ of the form $f'(y)\frac{dy}{dx} + P \cdot f(y) = Q$ — (1)

where P & Q are fⁿˢ of x or constants can be reduced to the linear form by putting

$f(y) = v$

$\therefore f'(y)\frac{dy}{dx} = \frac{dv}{dx}$ in eqⁿ (1).

$\therefore \frac{dv}{dx} + Pv = Q$ — (2)

Eqⁿ (2) is linear in v and can be solved using the method of L.D.E.

- Finally, substituting $v = f(y)$, we get the solⁿ of eqⁿ

## * Examples:-

(1) Solve $\sec^2 y \frac{dy}{dx} + x \tan y = x^3$

→

The given D.E is of the form

$f'(y)\frac{dy}{dx} + P f(y) = Q$.

$\therefore f(y) = \tan y$

Putting $\tan y = v$.

$\therefore \sec^2 y \frac{dy}{dx} = \frac{dv}{dx}$.

$\therefore$ The eqⁿ reduces to, $\frac{dv}{dx} + v \cdot x = x^3$

which is linear in v.

$\therefore$ I.F. $= e^{\int P\,dx} = e^{\int x\,dx} = e^{x^2/2}$.

$\therefore$ R.S. is $v(I.F.) = \int Q(I.F.)\,dx + C$

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$\therefore v\, e^{\frac{x^2}{2}} = \int x^3 e^{\frac{x^2}{2}} dx + c$

$\therefore e^{\frac{x^2}{2}} \tan y = e^{\frac{x^2}{2}} (x^2 - 2) + c.$

(2) $\frac{dy}{dx} + \frac{1}{x} = \frac{e^y}{x^2}.$

$\Rightarrow$

Dividing by $e^y$,

$e^{-y} \frac{dy}{dx} + e^{-y} \frac{1}{x} = \frac{1}{x^2}.$ — (1)

Let, $e^{-y} = v$

$\therefore e^{-y} \frac{dy}{dx} = \frac{dv}{dx}$

$\therefore$ From (1), $\frac{dv}{dx} + \frac{1}{x} v = \frac{1}{x^2}$ — (2)

Which is linear in $v$.

$\therefore P = \frac{1}{x}, \quad Q = \frac{1}{x^2}.$

I.F. $= e^{\int \frac{1}{x} dx} = e^{\log x} = x.$

$\therefore$ G.S. of eq$^n$ (2) is,

$x \cdot v = \int \frac{1}{x^2} x \, dx + c$

$= \int \frac{1}{x} dx + c$

$= \log x + c$

$\therefore x \cdot e^{-y} = \log x + c.$

(3) $\frac{dz}{dx} + \frac{z}{x} \log x = \frac{z}{x^2} (\log z)^2$

$\rightarrow$

Dividing by $z(\log z)^2$,

$\frac{1}{z(\log z)^2} \frac{dz}{dx} + \frac{1}{x \log z} = \frac{1}{x^2}$ — (1)

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Let, $\frac{-1}{\log z} = v$

$\therefore \frac{1}{(\log z)^2} \cdot \frac{1}{z} \frac{dz}{dx} = \frac{dv}{dx}$

$\therefore$ From (1),

$\frac{dv}{dx} - \frac{v}{x} = \frac{1}{x^2}$ ——— (2)

Which eq$^n$ is linear in $v$.

$P = \frac{-1}{x}$, $Q = \frac{1}{x^2}$.

I.F. $= e^{\int -\frac{1}{x} dx} = e^{-\log x} = e^{\log x^{-1}} = \frac{1}{x}$.

$\therefore$ G.S. of eq$^n$ (2) is

$\frac{1}{x} \cdot v = \int \frac{1}{x^2} \cdot \frac{1}{x} dx + C$

$= \int x^{-3} dx + C$

$\frac{1}{x}\left(\frac{-1}{\log z}\right) = \frac{x^{-2}}{-2} + C$

$\therefore \frac{1}{x \log z} = \frac{1}{2x^2} + C$

## * Exact Differential Equations:-

Any first order differential equation which is obtained by differentiation of its general solution without any elimination or reduction of terms is known as exact differential equation.

- If $f(x, y) = c$ is the general solution then

$df = 0$

$\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0$.

$M(x, y)\, dx + N(x, y)\, dy = 0$ ——— (1)

represents an exact differential equation.

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where, $M(x,y) = \frac{\partial f}{\partial x}$, $N(x,y) = \frac{\partial f}{\partial y}$.

$$\frac{\partial M}{\partial y} = \frac{\partial^2 f}{\partial y \partial x}, \quad \frac{\partial N}{\partial x} = \frac{\partial^2 f}{\partial x \partial y}.$$

But, $\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}$.

Therefore, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

Thus, the necessary condition for a differential equation to be an exact differential equation is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

- The solution of eq$^n$ (1) can be written as

$$\int_{y\text{-constant}} M(x,y)\,dx + \int (\text{terms of } N \text{ not containing } x)\,dy = c.$$

- Sometimes, integration of M w.r. to x is tedious whereas N can be integrated easily w.r. to y.
- In this case, the solution can be written as

$$\int (\text{terms of } M \text{ not containing } y)\,dx + \int_{x\text{ constant}} N(x,y)\,dy = c.$$

* **Examples:-**

① Solve $(x^3 + 3xy^2)\,dx + (3x^2y + y^3)\,dy = 0$

→

Here, $M = x^3 + 3xy^2$, $N = 3x^2y + y^3$.

$$\frac{\partial M}{\partial y} = 6xy, \quad \frac{\partial N}{\partial x} = 6xy$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

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- The eqⁿ is exact.

- The General solution is

$\int_{\text{y-constant}} M\,dx + \int$ (Terms of N not containing x) $dy = C$.

$\therefore \int (x^3 + 3xy^2)\,dx + \int y^3\,dy = C$

$\therefore \frac{x^4}{4} + \frac{3}{2}x^2 + \frac{y^4}{4} = C.$

(2) $\frac{dy}{dx} + \frac{y\cos x + \sin y + y}{\sin x + x\cos y + x} = 0.$

→

Rewritting the eqⁿ,

$\frac{dy}{dx} = -\frac{y\cos x + \sin y + y}{\sin x + x\cos y + x}$

$\therefore (\sin x + x\cos y + x)\,dy = -(y\cos x + \sin y + y)\,dx$

$\therefore (y\cos x + \sin y + y)\,dx + (\sin x + x\cos y + x)\,dy = 0$

Here, $M = y\cos x + \sin y + y$, $N = \sin x + x\cos y + x$.

$\frac{\partial M}{\partial y} = \cos x + \cos y + 1$, $\frac{\partial N}{\partial x} = \cos x + \cos y + 1$

Here, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

∴ The eqⁿ is exact.

Hence, The general solution is

$\int_{\text{y-constant}} M\,dx + \int$ (Terms of N not containing x) $dy = C$.

$\therefore \int (y\cos x + \sin y + y)\,dx + \int 0\,dy = C$

$\therefore y\sin x + x\sin y + xy = C.$

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(3) $[(x+1)e^x - e^y]\,dx - xe^y\,dy = 0$.

→

$M = (x+1)e^x - e^y$, $N = -xe^y$

$\frac{\partial M}{\partial y} = -e^y$, $\frac{\partial N}{\partial x} = -e^y$

Here, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

∴ The eq$^n$ is exact.

∴ The general solution is

$\int M\,dx + \int$ (Terms of N not containing x) $dy = c$.
y-constant

$\therefore \int [(x+1)e^x - e^y]\,dx + \int 0\,dy = c$.

$\therefore \int (x+1)e^x\,dx - \int e^y\,dx = c$

$\therefore (x+1)e^x - e^x - xe^y = c$

$\therefore xe^x - xe^y = c$

(4) For what values of a and b is the differential equation $(y+x^3)\,dx + (ax+by^3)\,dy = 0$ exact? Also, find the solution of the equation.

→

Here, $M = y+x^3$, $N = ax+by^3$

$\frac{\partial M}{\partial y} = 1$, $\frac{\partial N}{\partial x} = a$.

- The eq$^n$ will be exact if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

$\therefore 1 = a$.

∴ The eq$^n$ is exact for $a=1$ and for all values of b.

Substituting $a=1$ in the eq$^n$ $(y+x^3)\,dx + (x+by^3)\,dy = 0$ which is exact.

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- The general solution is

$\int M\,dx + \int$ (Terms of N not containing x) $dy = c$.
y-constant

$\therefore \int (y + x^3)\,dx + \int by^3\,dy = c.$

$\therefore xy + \frac{x^4}{4} + \frac{by^4}{4} = c.$

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